

Patrick Myers

Résumé

Department of Biomedical Engineering
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SUMMARY

Biomedical engineer with 7 years' experience in computational neuroscience – from developing closed-loop peripheral nerve stimulation systems to FDA-regulated commercial deployment. Core expertise in electrical nerve stimulation, neural signal processing, machine learning, and control theory. Experienced in all phases of translating research into clinical applications, including data collection via neurophysiological experiments, data analysis (EEG, iEEG, MRI, and LFP) and algorithm development in Python (Scikit-learn and PyTorch) and MATLAB, user-application development (REST API, AWS, front-end interfaces), and IP and regulatory application submissions. U.S. citizen, willing to relocate for in-person work.

EDUCATION

- 2023 – 2025 : **Doctor of Engineering**, Johns Hopkins University.
Dissertation: Restoration of Pain Processing from Neuropathic Chronic Pain: A Closed-Loop Neuromodulation Approach
- 2019 – 2019 : **Master of Engineering in Biomedical Engineering**,
Johns Hopkins University | GPA: 4.0/4.0
- 2015 - 2018 : **Bachelor of Science in Biomedical Engineering**,
Johns Hopkins University | GPA: 3.86/4.0
- Honors: Honors: Tau Beta Pi | Maryland Alpha Chapter, Alpha Eta Mu Beta | Johns Hopkins Chapter

EXPERIENCE

- 2023 –
PRESENT : **Graduate Research Scientist | Neuromedical Control Systems Lab**,
Johns Hopkins University, Advisors: Dr. Sridevi Sarma and Dr. Yun Guan
Closed-loop Control for the Restoration of Pain Processing in a Neuropathic Chronic Pain Condition

Developed an in-silico closed-loop peripheral nerve stimulation system to normalize pain processing in rats with neuropathic pain.

Performed in-vivo electrophysiological recordings to collect thalamic evoked potentials to noxious peripheral nerve stimuli in nerve injured and sham operated rats.

Designed constrained state-space models from the evoked potential recordings to provide a mechanistic interpretation of pain processing changes after injury and therapy.

Skills/Technologies: Python (neurophysiological data analysis), MATLAB (state space modeling, closed-loop controller design), Electrophysiological experiments in rodents

- 2019 – 2024 : **Director of Product Development | Neurologic Solutions**
Software start-up providing diagnostic support for potential Epilepsy patients

Translated technology to localize the seizure onset zone of medically refractory Epilepsy patients as an aid for pre-surgical planning.

Co-lead efforts for FDA 510K clearance, including performing Risk Analysis, documenting requirements and design, and designing usability, verification, and validation procedures.

Co-designed and built REST API and surrounding architecture for AWS powered cloud application in Python.

Skills/Technologies: REST API, Python (algorithm development), Java (front-end), AWS, FDA 510K

2021 – 2023 : **Assistant Research Scientist | Neuromedical Control Systems Lab**
Johns Hopkins University, Advisor: Dr. Sridevi Sarma

Predicting Epilepsy Diagnosis from Diagnostically Uninformative Scalp EEG Recordings

Collected EEG recordings from suspected epilepsy patients at 5 large clinical centers.

Designed a machine learning pipeline to extract network-based features from scalp EEG recordings to be used in a series of models to predict a patient's epilepsy status.

Ongoing prospective observational tests at 3 clinical centers with SBIR Phase II funding.

Skills/Technologies: Python (algorithm development), Machine Learning, MNE

PUBLICATIONS

Selected Publications

- 2024 : **Diagnosing Epilepsy with Normal Interictal EEG Using Dynamic Network Models**
Patrick Myers, Kristin M. Gunnarsdottir, Adam Li, ..., Sridevi V. Sarma and Khalil S. Husari, **Annals of Neurology**
- 2024 : **Neural Fragility of the Intracranial EEG Network Decreases after Surgical Resection of the Epileptogenic Zone**
Adam Li, Patrick Myers, ..., Sridevi V. Sarma, **In Preparation, medRxiv**
- 2021 : **Standardizing Human Brain Parcellations**
Ross M. Lawrence, Eric W. Bridgeford, Patrick E. Myers, ..., Joshua T. Vogelstein, **Nature Scientific Data**

Conference Proceedings

- 2024 : **Closed-Loop Peripheral Nerve Stimulation for the Restoration of Normal Pain Processing in Neuropathic Rats**
Patrick Myers, et al., **Conference on Decision and Control**
- 2024 : **Improving Diagnostic Yield of Scalp EEG Using Dynamic Network Modeling Analysis**
Patrick Myers, et al., **AI in Epilepsy Best Poster Award Winner**
- 2023 : **Using Dynamic Network Models of Routine Scalp EEG to Diagnose Epilepsy**
Patrick Myers, et al., **American Epilepsy Society**
- 2022 : **Diagnosing Epilepsy from Scalp EEG with and without IEDs: A Network-Based Analysis**
Patrick Myers, et al., **American Epilepsy Society**
- 2021 : **Neural Fragility as an iEEG Biomarker of the Seizure Onset Zone**
Patrick Myers, et al., **American Epilepsy Society**